SAM2-Only Training-Free Co-Saliency -- Full Project Report

Everything tried, what worked, what didn't -- with measured numbers. (as of 2026-07-03)

1. Goal and constraints

Find the common (co-salient) object across a group of images and segment it, using **SAM2 as the only model** (no DINO, SigLIP, CLIP, or any trained CoSOD network), fully **training-free**, with **category-agnostic** math (no constants tuned to a specific object class; scale-free percentile/rank thresholds only).

2. Base per-image pipeline (fixed structure throughout)

```
image (any res; logic at WORK_MAXSIDE=1280)
-> SAM2 encoder -> image_embed (64x64x256)
-> latent SNIC superpixels (clustering directly on the 256-d embedding map)
-> prompts (superpixel centroids, or uniform grid mode)
-> SAM2 object proposals -> NMS
-> subtraction into a partition: "clubs" (smallest object wins each pixel)
-> super-embeddings: mean image_embed per superpixel (L2-normed)
-> clubbed super-embeddings (size-weighted mean per club)
```

Earlier pipeline work (pre co-saliency): bleed-mask rejection (>25% of image), two-clock merge fix (03.jpg), bag+clock and smiley+clock separation (01/12.jpg), 4K/HD capability, latent SNIC replacing PCA+pysnic, SPAM assignment variants.

3. Co-saliency methods tried -- chronological ledger

#	Method / idea	Result (clock set)	Verdict
1	Inter-image consensus A-D (L2-norm -> P*P^T -> max per other image -> mean)	~13/16	baseline; background-prone
2	Background border penalty (Tip #1) + Otsu/top-k masks + vectorized matmul	~13/16	kept as stage-1 pieces
3	TF-SSD / IPS scoring (paper)	--	discarded ("forget IPS")
4	max instead of mean aggregation	~13/16	no change in ceiling
5	NxN block-matrix formulation	identical to flat (diff 2e-7)	equivalent, kept as viz
6	Greyscale-image variant	same ceiling	no advantage
7	Group-center proxy Fc + top-k prompt points (CoSOD paper)	~13-14/16	complementary errors
8	Ensemble consensus x proxy (soft-AND)	~14/16	kept idea
9	SAM2 refinement prompts: point / negative point / mask-prompt / mask+anchor	cleaner masks	mask+anchor best
10	Trained CoSOD (VGG16/ConvLSTM paper)	--	rejected (training-free constraint)
11	RBD boundary connectivity saliency	coherence 0.588->0.713, ~14/16	**adopted**
12	Combined consensus x RBD x proxy -> SAM2 refine	~14-15/16	best of the early era
13	Dual-stream fusion with SigLIP (score-level / concat)	~16/16 but **violates constraint**	rejected by constraint
14	Per-dimension decomposition (256 per-channel NxN matrices, pooled back to 256-d)	11-13/16	**worse** -- max(sum)<=sum(max): decoupling channels breaks same-object matching
15	Spectral consensus (principal eigenvector) + RBD foreground	13/16	kills backgrounds
16	Coverage / ubiquity (mean of worst-k per-image matches)	12/16 alone	re-admits backgrounds alone
17	fg x eigenvector x coverage combined	12-14/16	mug/teddy persist
18	Set-based min-Chamfer similarity (superpixel sets, min of two directions)	14/16	**kills uniform distractors** (cup, books)
19	Union candidates (connected club unions <=3)	15/16	**fixes fragmented objects**
20	Consensus assignment (coordinate ascent), gated, global clique + restarts	14-15/16	whack-a-mole among {00,07,11,14}; measured inversion: pillow 0.566 vs clock 0.550
21	Geometric layout consistency (part-position correlation)	pillow 0.54 > clock 0.35	**refuted by measurement**, discarded
22	Orphan-club repair (label-map regions with no superpixel mapping)	12->13 on color data	**bug fix, adopted**
23	Second-order profile similarity (candidates match the same things)	<=13/16	failed -- profiles dominated by generic signal

#	Method / idea	Result (clock set)	Verdict
24	tf-idf genericness weighting (multiplicity-based superpixel weights)	no change	no effect at any beta
25	Union internal-coherence gate	broke imgs 0/4/6	destructive -- real objects ARE dark+light unions
26	Marginal-gain prune of picked unions	fixed img14 -> 14/16	**adopted**
27	Annealed quantile agreement (worst-parts must match)	fixed 11,14; broke 00	circular; hysteresis window only 0.986-1.012, too thin
28	Frankenstein-union discovery (mug+blanket, pillow+sheet mimic dark-body+white-face)	diagnosis	key insight driving #29
29	**SAM2 crop re-encoding** of finalists (raw + masked, object scale)	raw 14/16, masked+ev 15/16	**the ceiling-breaker**
30	**Soft-AND consensus clique**: nrm(qsim)xnrm(raw)xnrm(masked)+lambda*evidence	**16/16**	**FINAL METHOD** -- ablation: drop any view -> 14/16
31	Robustness sweep	16/16 for q in {0.2,0.3,0.5} at lambda=0.15	lambda is the one balanced knob
32	Multi-scale super-embeddings (fuse high_res_feats 256^2+128^2)	clocks 16->13	**regression, reverted** -- hr feats are structural, not semantic
33	Multi-instance, relative threshold	pillow 102% > white clock 93%	failed -- weak winners inflate ratios
34	Multi-instance, absolute P25 bar (mean agreement)	white clock in; dotted mug also in (0.463 vs bar 0.456)	1 borderline false positive
35	**Multi-instance, top-3 near-duplicate rule** (P25 of winners' top-3)	white clock 0.627 [OK], mug 0.521 [X] (bar 0.550)	**adopted** -- 20% margin
36	Mask refinement: mask+box+anchor decoder prompt	crisp masks	adopted
37	**Zoom refinement** (decode on object crop; padded box)	bells/gear recovered; HD-sharp	**adopted**
38	Largest-connected-component + hole fill	speckles removed	adopted (cosmetic)
39	Grid prompts + MIN_AREA_FRAC=0.0005 for tiny objects	dice reachability 12->13/16	pipeline dial, adopted for tiny sets
40	Fine superpixels (BASE_N_SP 600)	recall unchanged	not the binding constraint
41	Size-stratified finalist shortlist (small_k)	clover/dice unchanged; clock 16/16 held	kept (harmless, principled)
42	Top-3 crop gate in expansion	clover unchanged	kept; no effect measured
43	Extent snapping (joint coordinate descent on pick extents, top-3 crops)	dice 2->2, clover 5->5, clock 16->16	**no net effect**; produced the context-near-duplicate diagnosis

4. Final method (v_final.py, K=10, q=0.3, lambda=0.15)

1. Candidates: clubs + connected unions <=3 (orphan-repaired)
2. Stage-1: min-Chamfer set similarity; evidence = RBD x eigenvector x robust coverage; gated assignment
3. Finalists: top-10 by evidence U top-10 by agreement U top-5 small-stratum
4. SAM2 crop re-encoding of every finalist (raw + masked) -> object-centric 256-d
5. Soft-AND consensus clique + multi-restart coordinate ascent
6. Marginal-gain prune + crop-gated expansion
7. Multi-instance: top-3 agreement >= P25(winners' top-3), non-overlapping
8. Zoom mask refinement (SAM2 decoder on object crop, padded box, LCC+fill)

5. Evaluation protocol

Hand-annotated GT boxes per image (verified visually); hit requires **union-purity >= 0.6** (pick lies on GT objects) AND **coverage >= 0.25** of at least one instance box. Plus visual verification of every reported number. Known criterion pitfall found and fixed: cluster boxes for sparse multi-instance sets made perfect picks "unreachable" (dice).

6. Results

Dataset	Strict score	Visual	Candidate-space ceiling	Notes
Alarm clocks (16)	**16/16**	16/16	16/16	+ 2nd instance in img03, no false extras
Airplanes HD (16 of 586 Airbus)	not box-annotated	**16/16**	--	zero-shot, frozen params, ~25 min
Clover (10)	5/10	~8/10 on-object	9/10	partial-leaf picks; img3 unreachable

Dataset	Strict score	Visual	Candidate-space ceiling	Notes
Dice (16)	2/16	~4/16	13/16 (grid prompts)	picks contain dice but absorb context

7. Characterized ceilings (measured, not speculated)

1. **Appearance ≠ category**: SAM2 embeddings encode appearance. Where a look-alike matches the group better than a weak true instance (pillow 0.566 vs clock 0.550), no training-free inference re-orders it.

2. **Dice / context co-recurrence**: the group's *surfaces* (felt, boards, cards) are near-duplicates across images (top-3 crop agreements 0.87-0.95) -- preferring dice over surfaces is a semantic judgment appearance cannot make.

3. **Clover / heterogeneous subgroups**: dark-center vs pure-green clovers -> consensus keeps only the universally common part; partial picks validate each other (self-consistent wrong extent). Five independent mechanisms failed identically.

4. **Tiny-object recall**: objects <= ~1-2% of image area dissolve in the proposal->partition stage; grid prompts + small min-area recover some (dice 12->13 reachable), motion-blurred / sub-40px objects stay unreachable.

5. **Decoder resolution**: fixed by zoom refinement (decode on crop); residual limit is SAM2's 1024-px internal budget per crop.

8. What is honestly achieved vs not

Achieved: clocks 16/16 (strict + visual, regression-proof), airplanes 16/16 (visual, zero-shot), principled multi-instance recovery, HD-crisp masks, no hand constants, SAM2-only throughout, one-command runner (`variants/run_dataset.py`).

Not achieved: clover beyond 5/10 strict, dice beyond 2/16, clock-07 selection purity (blanket club), standard benchmark numbers (CoCA/CoSOD3k never run -- no SOTA claim possible), group-size scaling beyond N~16 (O(N^2)).

9. Assets

- Code: `/Users/arjunsingla/sam2/` (pipeline_final.py, snic_latent.py) and `/Users/arjunsingla/sam2/variants/` (harness, v_base, v_combo, v_anneal, v_reencode, v_final, multi_instance, refine_masks, extent_snap, run_dataset, step_run, gt_color/gt_dice/gt_clover)

- Data: `/Users/arjunsingla/sam2/data/*.npy` (durable; /tmp gets wiped)

- Renders: `/Users/arjunsingla/Desktop/SAM2_Pipeline/cosaliency/` (FINAL_16of16, FINAL_refined, RESULT_airbus16, GENERAL_clover, GENERAL_dice2, gt box overlays, diagnostics)

10. Open next steps

1. CoCA / CoSOD3k benchmark harness -> real standing vs published training-free methods

2. Dice/clover ceilings: would require semantic signal (excluded by constraint) or a group-prior beyond appearance

3. Batch runner for the remaining 570 Airbus images; mixed-type group stress test (Airbus+F16+C130)